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Thabo owns exactly 140 books, and each book is either paperback fiction, paperback nonfiction, or hardcover nonfiction. If he owns 20 more paperback nonfiction books than hardcover nonfiction books, and twice as many paperback fiction books as paperback nonfiction books, how many hardcover nonfiction books does Thabo own?

A. 10

B. 20

C. 30

D. 40

E. 50

Algebra Simultaneous First-Degree Equations

Let F represent the number of paperback fiction books that Thabo owns; N_p, the number of paperback nonfiction books; and N_h, the number of hardcover nonfiction books. It is given that $F + N_p + N_h = 140$, $N_p = N_h + 20$, and $F = 2N_p = 2(N_h + 20)$. It follows that

$$\begin{aligned} F + N_p + N_h &= 140 \text{ given} \\ 2(N_h + 20) + (N_h + 20) + N_h &= 140 \text{ by substitution} \\ 4N_h + 60 &= 140 \text{ combine like terms} \\ 4N_h &= 80 \text{ subtract 60 from both sides} \\ N_h &= 20 \text{ divide both sides by 4} \end{aligned}$$

The correct answer is B.

•

What is 35 percent of the sum of 1.4 and $\frac{1}{5}$?

A. 0.42

B. 0.56

C. 0.85

D. 1.55

E. 1.95

Arithmetic Applied Problems

Thirty-five percent of the sum of 1.4 and $\frac{1}{5}$ is 0.35(1.4 + 0.2) = 0.35(1.6) = 0.56.

The correct answer is B.

•

If $x > 0$, $\frac{x}{50} + \frac{x}{25}$ is what percent of x?

A. 6%

B. 25%

C. 37%

D. 60%

E. 75%

Algebra Arithmetic; Simplifying Algebraic Expressions; Percents

Because we want a percent, use a common denominator of 100 to combine the two terms.

$\frac{x}{50} + \frac{x}{25} = \frac{2x}{100} + \frac{4x}{100} = \frac{6x}{100} = \left(\frac{6}{100}\right)x$, which is 6% of x.

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$\frac{}{50} + \frac{}{25} = \frac{}{100} + \frac{}{100} = \frac{}{100} = \left(\frac{}{100}\right)x$, which is 6% of x.

The correct answer is A.

•

During a certain time period, Car X traveled north along a straight road at a constant rate of 1 mile per minute and used fuel at a constant rate of 5 gallons every 2 hours. During this time period, if Car X used exactly 3.75 gallons of fuel, how many miles did Car X travel?

A. 36

B. 37.5

C. 40

D. 80

E. 90

Arithmetic Applied Problems

The car used fuel at the rate of 5 gallons every 2 hours, and thus after using 3.75 gallons of fuel, the car had traveled for $\left(3.75\right)\left(\frac{2}{5}\right)$ hours. One way to obtain this value is indicated by the calculations in the following table.

5 gal	corresponds to	2 hr	Given
1 gal	corresponds to	$\frac{2}{5}\text{hr}$	Divide by 5
3.75 gal	corresponds to	$\left(3.75\right)\left(\frac{2}{5}\right)\text{hr}$	Multiply by 3.75

Since $(3.75)\left(\frac{2}{5}\right) = \left(\frac{15}{4}\right)\left(\frac{2}{5}\right) = \frac{3}{2}\text{hours} = 90$

minutes, the car drove for 90 minutes at a rate of 1 mile per minute. Therefore, the total distance the car drove was

$\text{distance} = \text{rate} \times \text{time} = \left(1\frac{\text{mi}}{\text{min}}\right)(90\text{ min}) = 90\text{ miles}.$

The correct answer is E.

•

Which of the following is closest to the value of $\frac{999}{100 + \frac{1}{999}}$

A. 10

B. 1

C. 0.1

D. 0.01

E. 0.001

Arithmetic Estimation

$\frac{999}{100 + \frac{1}{999}}$ is close to $\frac{1000}{100} = 10.$

The correct answer is A.

•

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The population of City X is 50 percent of the population of City Y. The population of City X is what percent of the total population of City X and City Y?

A. 25%

B. $33\frac{1}{3}\%$

C. 40%

D. 50%

E. $66\frac{2}{3}\%$

Arithmetic Percents

Let x be the population of City X and y be the population of City Y. Since $x = 0.5y$, then $y = 2x$. So, the total population of City X and City Y is $x + y = x + 2x = 3x$. Thus, x is $\frac{1}{3}$, or $33\frac{1}{3}\%$, of $x + y$.

The correct answer is B.

•

A bicycle store purchased two bicycles, one for $250 and the other for $375, and sold both bicycles at a total gross profit of $250. If the store sold one of the bicycles for $450, which of the following could be the store's gross profit from the sale of the other bicycle?

A. $75

B. $100

C. $125

D. $150

E. $175

Arithmetic Profit and Loss

Let $450 and $\$R$ be the individual revenues from selling the two bicycles. Then the unknown profit from selling one of the bicycles is either $\$R - \$250$ or $\$R - \$375$.

$$\begin{aligned} \text{total profit} &= (\textit{total}\ \text{revenue}) - (\text{total cost}) \\ \$250 &= (\$450 + \$R) - (\$250 + \$375) \\ \$250 &= \$R - \$175 \\ \$425 &= \$R \end{aligned}$$

From this it follows that the unknown profit is either $425 – $250 = $175 or $425 – $375 = $50.

The correct answer is E.

•

If $k^2 = m^2$, which of the following must be true?

A. $k = m$

B. $k = -m$

C. $k = |m|$

D. $k = -|m|$

E. $|k| = |m|$

Algebra Simplifying Algebraic Expressions

One method of solving this problem is to take the nonnegative square root of both sides of the equation $k^2 = m^2$ and then since $\sqrt{u^2} = |u|$ for all real numbers u, it follows that $|k| = \sqrt{k^2} = \sqrt{m^2} = |m|$.

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The table below shows that each of the other answer choices can be true but can also be false. For each pair of values for k and m, it is true that $k^2 = m^2$.

	True for		False for	
	k	m	k	m
A $k = m$	3	3	3	–3
B $k = -m$	–3	3	3	3
C $k = \|m\|$	3	–3	–3	–3
D $k = -\|m\|$	–3	–3	3	–3

The correct answer is E.

•

Makoto, Nishi, and Ozuro were paid a total of $780 for waxing the floors at their school. Each was paid in proportion to the number of hours he or she worked. If Makoto worked 15 hours, Nishi worked 20 hours, and Ozuro worked 30 hours, how much was Makoto paid?

A. $52

B. $117

C. $130

D. $180

E. $234

Arithmetic Ratio and Proportion

Makoto, Nishi, and Ozuro worked a total of 15 + 20 + 30 = 65 hours and were paid a total of $780. Since Makoto worked 15 of the 65 hours and he was paid in proportion to the number of hours he worked, his pay was $\frac{15}{65}(\$780)$.

To determine which of the answer choices equals this amount of money, note that $\frac{15}{65}$ can be reduced to $\frac{3}{13}$ by dividing the numerator and denominator by 5. Also note that 780 can be factored as (10)(78) = (10)(2)(39) = (10)(2)(3)(13). Then calculate $\frac{3}{\cancel{13}}(10)(2)(3)(\cancel{13}) = \left(3\right)\left(10\right)\left(2\right)\left(3\right) = 180.$

Tip: In problems requiring calculations with fractions, it can be helpful to reduce fractions and utilize cancellation as much as possible.

The correct answer is D.

•

If x is a positive integer and $4^x - 3 = y$, which of the following CANNOT be a value of y?

A. 1

B. 7

C. 13

D. 61

E. 253

Arithmetic Exponents

This can be solved by calculating the value of $4^x - 3$ for the first few positive integer values of x.

x	4^x	$4^x - 3$	answer choice
1	4	1	A
2	16	13	C

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3	64	61	D
4	256	253	E

Alternatively, this can be solved by observing that 4^x always has units digit 4 or 6—the product of two integers with units digit 4 has units digit 6, the product of two integers with units digit 6 has units digit 4, etc.—and therefore any integer that does not have units digit 4 – 3 = 1 or 6 – 3 = 3 cannot be the value of $4^x - 3$ for some positive integer value of x.

The correct answer is B.

•

The regular price per can of a certain brand of soda is $0.40. If the regular price per can is discounted 15 percent when the soda is purchased in 24-can cases, what is the price of 72 cans of this brand of soda purchased in 24-can cases?

A. $16.32

B. $18.00

C. $21.60

D. $24.48

E. $28.80

Arithmetic Percents

The price per can with the 15% discount is 100% – 15% = 85% of the regular price of 40 cents per can; that is, the discounted price is 0.85(40) = 34 cents per can. So, 72 cans at 34 cents apiece cost 72 × 34 cents = 2,448 cents, or $24.48.

The correct answer is D.

•

The quotient when a certain number is divided by $\frac{2}{3}$ is $\frac{9}{2}$.

What is the number?

A. $\frac{4}{27}$

B. $\frac{1}{3}$

C. 3

D. 6

E. $\frac{27}{4}$

Arithmetic Operations with Rational Numbers

Let N be the unknown number. The quotient when N is divided by $\frac{2}{3}$ is $N \div \frac{2}{3}$, which equals $N \times \frac{3}{2} = \frac{3}{2}N$ (invert and multiply). Since this quotient equals $\frac{9}{2}$, it follows that $\frac{3}{2}N = \frac{9}{2}$. Therefore, $N = \left(\frac{9}{2}\right)\left(\frac{2}{3}\right) = 3$.

Tip: Sometimes a correct method of calculation will be intuitively evident if difficult-to-conceptualize numbers are replaced with easy-to-conceptualize numbers. For example, if the first sentence had instead been "The quotient when a certain number is divided by 2 is 3" (i.e., after dividing a certain number by 2, the result is 3), then it may be intuitively evident that the number can be obtained by multiplying 2 and 3. Thus, for this problem, the number can be obtained by multiplying $\frac{2}{3}$ and $\frac{9}{2}$.

The correct answer is C.

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The correct answer is C.

•

The price of gasoline at a service station increased from $1.65 per gallon last week to $1.82 per gallon this week. Sally paid $26.40 for gasoline last week at the station. How much more will Sally pay this week at the station for the same amount of gasoline?

A. $1.70

B. $2.55

C. $2.64

D. $2.72

E. $2.90

Arithmetic Applied Problems

Since $1.82 – $1.65 = $0.17, Sally paid $0.17 per gallon more this week than last week. Therefore, the total additional amount that Sally paid this week over last week is the number of gallons Sally purchased times $0.17.

Number of gallons purchased last week:

$$\frac{\text{total paid last week}}{\text{price per gallon last week}} = \frac{26.40}{1.65}$$

Number of gallons purchased this week: $\frac{26.40}{1.65}$

Total additional amount paid this week: $(\$0.17)\frac{26.40}{1.65} = (\$0.17)(16) = \$2.72.$

One way to lessen the arithmetic computations is to work with integers and factor.

$$0.17\left(\frac{26.4}{1.65}\right) = \frac{(17)(264)}{1,650}$$ multiply numerator and denominator by 1,000 for ease of calculation

$$= \frac{(17)(3 \times 8 \times 11)}{3 \times 5 \times 11 \times 10}$$ factor

$$= \frac{(17)(\cancel{3} \times 8 \times \cancel{11})}{\cancel{3} \times 5 \times \cancel{11} \times 10}$$ identify common factors

$$= \frac{17 \times 8}{5 \times 10}$$ cancel common factors

$$= \frac{136}{50}$$ multiply

$$= \frac{13.6}{5}$$ divide numerator and denominator by 10

$$= 2.72$$ divide

The correct answer is D.

Monthly Charge for Low-Use Telephone Contract Offered by Company X

Monthly rate (up to 75 message units)	20% less than standard rate of $10.00
Per unit in excess of 75 message units	$0.065

•

Harry spent 25% of his $2,200 May take-home pay on rent and 30% of the remainder on food. How much of his May take-home pay did Harry spend on food?

A. $165

B. $495

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B. $495

C. $550

D. $660

E. $1,045

Arithmetic Percents

Since Harry spent 25% of the $2,200 May take-home pay on rent, the remainder was 75% × $2,200 = $1,650. So, of his May take-home pay, Harry spent 30% × $1,650 = $495 on food.

The correct answer is B.

•

If $2x + y = 7$ and $x + 2y = 5$, then $\frac{x+y}{3} =$

A. 1

B. $\frac{4}{3}$

C. $\frac{17}{5}$

D. $\frac{18}{5}$

E. 4

Algebra Simultaneous Equations

Adding the equations $2x + y = 7$ and $x + 2y = 5$ gives $3x + 3y = 12$, or $x + y = 4$. Dividing both sides of the last equation by 3 gives $\frac{x+y}{3} = \frac{4}{3}$.

The correct answer is B.

•

City X has a population 4 times as great as the population of City Y, which has a population twice as great as the population of City Z. What is the ratio of the population of City X to the population of City Z?

A. 1:8

B. 1:4

C. 2:1

D. 4:1

E. 8:1

Arithmetic Ratio and Proportion

Let X, Y, and Z be the populations of Cities X, Y, and Z, respectively. It is given that $X = 4Y$, and $Y = 2Z$ or $Z = \frac{Y}{2}$. Then, $\frac{X}{Z} = \frac{4Y}{\frac{Y}{2}} = (4Y)\left(\frac{2}{Y}\right) = \frac{8}{1}$.

The correct answer is E.

Total Membership of Organization X, 1988

Emeritus	78
Fellows	9,209
Partners	35,509
Associates	27,909

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Affiliates	2,372

•

According to the table above, the number of fellows was approximately what percent of the total membership of Organization X? **Note: All categories of membership are mutually exclusive.**

A. 9%

B. 12%

C. 18%

D. 25%

E. 35%

Arithmetic Percents

Since no one is in more than one of the categories listed in the table, the total membership is the sum of the numbers in the table. We're just asked to find an approximate percentage, so let's round each number to the nearest thousand and then divide each by a thousand. Thus, we find that the approximate percentage of the total membership who are fellows was $100\% \times \frac{9}{(9+36+28+2)} = 100\% \times \frac{9}{75} = 12\%$.

The correct answer is B.

•

If $S = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \frac{1}{9^2} + \frac{1}{10^2}$,

which of the following is true?

A. $S > 3$

B. $S = 3$

C. $2 < S < 3$

D. $S = 2$

E. $S < 2$

Arithmetic Estimation

Because $\frac{1}{3^2}$ is less than $\frac{1}{2^2}$ and each of $\frac{1}{5^2}, \frac{1}{6^2}, \frac{1}{7^2}, \frac{1}{8^2}, \frac{1}{9^2}$, and $\frac{1}{10^2}$ is less than $\frac{1}{4^2}$,

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \frac{1}{9^2} + \frac{1}{10^2}$$

$$< 1 + 2\left(\frac{1}{2^2}\right) = 1\frac{1}{2} + 7\left(\frac{1}{4^2}\right) = \frac{7}{16}$$

Therefore, $S < 1\frac{1}{2} + \frac{7}{16} < 2$.

The correct answer is E.

•

A manufacturer of a certain product can expect that between 0.3 percent and 0.5 percent of the units manufactured will be defective. If the retail price is $2,500 per unit and the manufacturer offers a full refund for defective units, how much money can the manufacturer expect to need to cover the refunds on 20,000 units?

A. Between $15,000 and $25,000

B. Between $30,000 and $50,000

C. Between $60,000 and $100,000

D. Between $150,000 and $250,000

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D. Between $150,000 and $250,000

E. Between $300,000 and $500,000

Arithmetic Applied Problems

The expected number of defective units is between 0.3% and 0.5% of 20,000, or between (0.003)(20,000) = 60 and (0.005)(20,000) = 100. Since each unit has a retail price of $2,500, the amount of money needed to cover the refunds for the expected number of defective units is between 60($2,500) and 100($2,500), or between $150,000 and $250,000.

The correct answer is D.

•

Which of the following is closest to $\sqrt{\frac{4.2(1,590)}{15.7}}$?

A. 20

B. 40

C. 60

D. 80

E. 100

Arithmetic Estimation

Replace the numbers that appear with nearly equal values that are perfect squares and then evaluate the resulting expression:

$$\sqrt{\frac{4\,(1,600)}{16}} = \sqrt{\frac{4 \times \cancel{16} \times 100}{\cancel{16}}} = \sqrt{4 \times 100}$$
$$= \sqrt{4} \times \sqrt{100} = 2 \times 10 = 20$$

The correct answer is A.

•

What is the value of $\sqrt{17} + \sqrt{47}$, to the nearest whole number?

A. 8

B. 11

C. 12

D. 15

E. 32

Arithmetic Equations

$\sqrt{17}$ is slightly greater than $\sqrt{16} = 4$, while $\sqrt{47}$ is slightly less than $\sqrt{49} = 7$. Thus, $\sqrt{17} + \sqrt{47} \approx 4 + 7 = 11$.

The correct answer is B.

•

Last week Chris earned x dollars per hour for the first 40 hours worked plus 22 dollars per hour for each hour worked beyond 40 hours. If last week Chris earned a total of 816 dollars by working 48 hours, what is the value of x?

A. 13

B. 14

C. 15

D. 16

E. 17

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E. 17

Algebra Applied Problems

Chris worked 40 hours at a rate of $\$x$ per hour, 48 – 40 = 8 hours at a rate of $22 per hour, and earned a total of $816.

$$\begin{aligned} 40x + 8(22) &= 816 \text{ given information} \\ 40x + 176 &= 816 \text{ multiply 8 and 22} \\ 40x &= 640 \text{ subtract 176 from both sides} \\ x &= 16 \text{ divide both sides by 40} \end{aligned}$$

The correct answer is D.

•

The value of $\dfrac{\frac{7}{8}+\frac{1}{9}}{\frac{1}{2}}$ is closest to which of the following?

A. 2

B. $\frac{3}{2}$

C. 1

D. $\frac{1}{2}$

E. 0

Arithmetic Estimation

First, note that division by $\frac{1}{2}$ is the same as multiplication by $\frac{2}{1}$ (invert and multiply), so, $\left(\frac{7}{8}+\frac{1}{9}\right) \div \frac{1}{2} = \left(\frac{7}{8}+\frac{1}{9}\right) \times \frac{2}{1} = \left(\frac{7}{8}+\frac{1}{9}\right) \times 2$. Then, because the question asks for the value "closest to" and not the exact value, estimate $\left(\frac{7}{8}+\frac{1}{9}\right) \times 2 \approx \left(\frac{7}{8}+\frac{1}{8}\right) \times 2$ and calculate $\left(\frac{7+1}{8}\right) \times 2 = \frac{8}{8} \times 2 = 2$.

Alternatively, calculate $\frac{7}{8}+\frac{1}{9}$ using the common denominator of 72, $\frac{7}{8}+\frac{1}{9} = \frac{63}{72}+\frac{8}{72} = \frac{71}{72}$, then estimate $\frac{71}{72} \approx \frac{72}{72} = 1$, then calculate $1(2) = 2$.

The correct answer is A.

•

The positive two-digit integers x and y have the same digits, but in reverse order. Which of the following must be a factor of $x + y$?

A. 6

B. 9

C. 10

D. 11

E. 14

Arithmetic Properties of Numbers

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Let m and n be digits. If $x = 10m + n$, then $y = 10n + m$. Adding x and y gives $x + y = (10m + n) + (10n + m) = 11m + 11n = 11(m + n)$, and therefore 11 is a factor of $x + y$.

The correct answer is D.

•

In a certain sequence of 8 numbers, each number after the first is 1 more than the previous number. If the first number is –5, how many of the numbers in the sequence are positive?

A. None

B. One

C. Two

D. Three

E. Four

Arithmetic Sequences

The sequence consists of eight consecutive integers beginning with –5:

–5, –4, –3, –2, –1, 0, 1, 2

In this sequence exactly two of the numbers are positive.

The correct answer is C.

•

A total of s oranges are to be packaged in boxes that will hold r oranges each, with no oranges left over. When n of these boxes have been completely filled, what is the number of boxes that remain to be filled?

A. $s - nr$

B. $s - \frac{n}{r}$

C. $rs - n$

D. $\frac{s}{n} - r$

E. $\frac{s}{r} - n$

Algebra Algebraic Expressions

If s oranges are packed r oranges to a box with no oranges left over, then the number of boxes that will be filled is $\frac{s}{r}$.

If n of these boxes are already filled, then $\frac{s}{r} - n$ boxes remain to be filled.

The correct answer is E.

•

If $0 < a < b < c$, which of the following statements must be true?

I. $2a > b + c$

II. $c - a > b - a$

III. $\frac{c}{a} < \frac{b}{a}$

A. I only

B. II only

C. III only

D. I and II

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E. II and III

Algebra Inequalities

Given $0 < a < b < c$, Statement I is not necessarily true. If, for example, $a = 1$, $b = 2$, and $c = 3$, then $0 < a < b < c$, but $2a = 2(1) < 2 + 3 = b + c$.

Given $0 < a < b < c$, then $c > b$, and subtracting a from both sides gives $c - a > b - a$. Therefore, Statement II is true.

Given $0 < a < b < c$, Statement III is not necessarily true. If, for example, $a = 1$, $b = 2$, and $c = 3$, then $0 < a < b < c$, but $\frac{c}{a} = \frac{3}{1} > \frac{2}{1} = \frac{b}{a}$.

The correct answer is B.

	Monday	Tuesday	Wednesday	Thursday
Company A	45	55	50	50
Company B	10	30	30	10
Company C	34	28	28	30
Company D	39	42	41	38
Company E	50	60	60	70

•

The table shows the numbers of packages shipped daily by each of five companies during a 4-day period. The standard deviation of the numbers of packages shipped daily during the period was greatest for which of the five companies?

A. A

B. B

C. C

D. D

E. E

Arithmetic Statistics

Since the standard deviation of a data set is a measure of how widely the data are scattered about their mean, find the mean number of packages shipped by each company and then calculate the deviations of the company's data from its mean.

For Company A, the mean number of packages shipped is $\frac{45 + 55 + 2(50)}{4} = 50$ and the deviations from this mean are −5, 5, 0, and 0. Note that the numbers of packages allow for quick calculations of the means. For example, for Company A the total number of packages is the sum of 45 + 55 = 100 and 50 + 50 = 100, and for Company D the total number of packages is the sum of 39 + 41 = 80 and 42 + 38 = 80.

The following table shows the means and deviations for the 5 companies.

	mean	deviations from mean
Company A	50	−5, 5, 0, 0
Company B	**20**	**−10, 10, 10, −10**
Company C	30	4, −2, −2, 0
Company D	40	−1, 2, 1, −2
Company E	60	−10, 0, 0, 10

This table shows that the magnitude of each of the 4 deviations for Company B is greater than or equal to the magnitude of any of the other deviations, which strongly suggests that the standard deviation is greatest for Company B.

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B.

For those interested, the standard deviations can be calculated as $\sqrt{\frac{\text{sum of (deviations)}^2}{4}}$. However, since we only wish to determine which company has the greatest standard deviation, it suffices to calculate the values of "sum of (deviations)^2", because dividing these values by 4 followed by taking a square root will not change their order (that is, the greatest will still be the greatest).

	(deviations)^2	sum of (deviations)^2
Company A	25, 25, 0, 0	50
Company B	**100, 100, 100, 100**	**400**
Company C	16, 4, 4, 0	24
Company D	1, 4, 1, 4	10
Company E	100, 0, 0, 100	200

The correct answer is B.

•

Company Q plans to make a new product next year and sell each unit of this new product at a selling price of $2. The variable costs per unit in each production run are estimated to be 40% of the selling price, and the fixed costs for each production run are estimated to be $5,040. Based on these estimated costs, how many units of the new product will Company Q need to make and sell in order for their revenue to equal their total costs for each production run?

A. 4,200

B. 3,150

C. 2,520

D. 2,100

E. 1,800

Algebra Applied Problems

Let x be the desired number of units to be sold at a price of $2 each. Then the revenue for selling these units is $\$2x$, and the total cost for selling these units is $(40\%)(\$2.00)x = \$0.80x$ plus a fixed cost of $5,040.

revenue = total cost	given requirement
$2x = 0.8x + 5,040$	given information
$1.2x = 5,040$	subtract $0.8x$ from both sides
$x = 4,200$	divide both sides by 1.2

The correct answer is A.

•

A small business invests $9,900 in equipment to produce a product. Each unit of the product costs $0.65 to produce and is sold for $1.20. How many units of the product must be sold before the revenue received equals the total expense of production, including the initial investment in equipment?

A. 12,000

B. 14,500

C. 15,230

D. 18,000

E. 20,000

Arithmetic Rate

Let n be the number of units desired. Then, in dollars, the revenue received is $1.2n$ and the total expense is $9,900 + 0.65n$. These two amounts are equal when $1.2n = 9,900 + 0.65n$, or $0.55n = 9,900$. Therefore

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$9,900 + 0.65n$. These two amounts are equal when $1.2n = 9,900 + 0.65n$, or $0.55n = 9,900$. Therefore, $n = \frac{9,900}{0.55} = \frac{990,000}{55} = \frac{90,000}{5} = 18,000$.

The correct answer is D.

Estimated Number of Home-Schooled Students by State, January 2001

State	Number (in thousands)
A	181
B	125
C	103
D	79
E	72

•

According to the table shown, the estimated number of home-schooled students in State A is approximately what percent greater than the number in State D?

A. 25%

B. 55%

C. 100%

D. 125%

E. 155%

Arithmetic Percents

The percent increase from the number in State D to the number in State A is

$$= \left(\frac{181,000 - 79,000}{79,000} \times 100\right)\% \quad \text{expression for percent increase}$$

$$= \left(\frac{181 - 79}{79} \times 100\right)\% \quad \text{reduce fraction}$$

$$= \left(\frac{102}{79} \times 100\right)\% \quad \text{subtract}$$

$$\approx \left(\frac{100}{80} \times 100\right)\% \quad \text{approximate}$$

$$= (1.25 \times 100)\% \quad \text{divide}$$

$$= 125\% \quad \text{multiply}$$

The correct answer is D.

•

When n liters of fuel were added to a tank that was already $\frac{1}{3}$ full, the tank was filled to $\frac{7}{9}$ of its capacity. In terms of n, what is the capacity of the tank, in liters?

A. $\frac{10}{9}n$

B. $\frac{4}{3}n$

C. $\frac{3}{2}n$

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D. $\frac{9}{4}n$

E. $\frac{7}{3}n$

Algebra Applied Problems

Let C represent the capacity of the tank, in liters. It follows that

$$\begin{aligned} \frac{1}{3}C + n &= \frac{7}{9}C && \text{given} \\ n &= \frac{7}{9}C - \frac{1}{3}C && \text{subtract } \frac{1}{3}C \text{ from both sides} \\ n &= \frac{4}{9}C && \text{combine like terms} \\ \frac{9}{4}n &= C && \text{divide both sides by } \frac{4}{9} \end{aligned}$$

The correct answer is D.

•

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